Periodic Approximation of Literal Multipliers on Determinant-Two Möbius Fibers

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Abstract

The determinant-two corridor of TPC-149 bounds every bounded periodic multiplier on a literal two-Möbius core, uniformly outside one exceptional set. We give the exact interface from that theorem to an arbitrary literal multiplier. The resulting bound is the sum of the periodic-core saving and a normalized L^1 approximation error. The unpenalized error term separates over residue fibers; its quadratic surrogate is solved by fiber means. The supremum-norm penalty in the full cost still couples those fibers. This is a positive actual-core weight-interface theorem. It is not yet a theorem for the missing physical occurrence registry, a generic phase or all prefixes, and its fixed-X-power exponent is zero.

Keywords. Möbius correlation; determinant two; periodic approximation; literal multiplier; exceptional scales.

Claim boundary. The theorem below acts on the literal determinant-two Möbius core once a multiplier w is supplied. It does not construct the frontier occurrence lift or certify that the unknown physical multiplier has small approximation cost. A logarithmic saving is not a positive fixed-X-power saving and cannot pay the $1/400$ endpoint. No prime-pair lower bound or twin-prime theorem is proved.

1 The inherited periodic corridor

Fix integers

$$a, s, R \geq 1, \qquad d, u \in \mathbb{Z}, \qquad (a, s) = 1, \qquad as \text{ odd}, \qquad su - ad = 2,$$

and put

$$q = as, \qquad t(z) = ad + qz, \qquad c_z = \mu(d + sz)\mu(u + az),$$
$$I_N = \{z \in \mathbb{Z} : N < t(z) \leq 2N\}.$$

TPC-149 proves the following consequence of the quantitative affine correlation theorem of Tao and Teräväinen [1].

Theorem 1.1 (TPC-149 periodic corridor). *There are absolute $\eta_0, \kappa_0 > 0$ such that for every sufficiently large X there is $\mathcal{E}_X^\star \subset [\sqrt{X}, X]$ satisfying*

$$\frac{1}{\log X} \int_{\mathcal{E}_X^\star} \frac{dt}{t} \ll (\log X)^{-\kappa_0}.$$

If $qR \leq (\log X)^{\eta_0}$ and $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X^\star$, then, simultaneously for every period-R function ρ,

$$\frac{q}{N} \left| \sum_{z \in I_N} c_z \rho(z) \right| \ll \|\rho\|_\infty (\log X)^{-\kappa_0}. \tag{1}$$

The word "simultaneously" is essential: choosing an approximant after seeing w does not require a union over its values.

2 The literal-weight interface

Definition 2.1 (Periodic approximation cost)**.** For $w : I_N \to \mathbb{C}$, define

$$\mathfrak{A}_{R,N}(w) = \inf_{\substack{\rho:\mathbb{Z}\to\mathbb{C} \\ \rho(z+R)=\rho(z)}} \left\{ \|\rho\|_\infty (\log X)^{-\kappa_0} + \frac{q}{N} \sum_{z\in I_N} |w(z) - \rho(z)| \right\}. \tag{2}$$

Theorem 2.2 (Literal-weight periodic approximation)**.** *Under the hypotheses of theorem 1.1,*

$$\boxed{\frac{q}{N} \left| \sum_{z\in I_N} c_z w(z) \right| \ll \mathfrak{A}_{R,N}(w).} \tag{3}$$

The exceptional set is exactly the one in theorem 1.1.

Proof. For any admissible ρ,

$$\sum_{I_N} c_z w(z) = \sum_{I_N} c_z \rho(z) + \sum_{I_N} c_z \{w(z) - \rho(z)\}.$$

The first term is controlled by (1). Since $|c_z| \le 1$, the normalized modulus of the second is at most

$$\frac{q}{N} \sum_{z\in I_N} |w(z) - \rho(z)|.$$

Take the infimum. No exceptional-set operation occurs. □

Corollary 2.3 (A usable promotion rule)**.** *Suppose an independently source-locked physical registry supplies $w = w_X$, a period R_X with $qR_X \le (\log X)^{\eta_0}$, and an approximant satisfying*

$$\|\rho_X\|_\infty \ll 1, \qquad \frac{q}{N} \sum_{I_N} |w_X - \rho_X| \ll (\log X)^{-\gamma}.$$

Then the weighted actual core has a logarithmic saving with exponent $\min\{\kappa_0, \gamma\}$*.*

Remark 2.4. The corollary is an implication, not a claim that the current frontier archive supplies w_X. That occurrence-level input remains unavailable.

3 Residue-fiber error optimization

For $r \in \mathbb{Z}/R\mathbb{Z}$, write

$$I_{N,r} = \{z \in I_N : z \equiv r \pmod{R}\}.$$

Proposition 3.1 (Exact separation)**.** *For every w,*

$$\inf_{\rho \text{ period } R} \sum_{I_N} |w(z) - \rho(z)| = \sum_{r \bmod R} \inf_{\zeta\in\mathbb{C}} \sum_{z\in I_{N,r}} |w(z) - \zeta|.$$

For the squared error, the unique minimizing value on every nonempty fiber is

$$\bar{w}_r = \frac{1}{|I_{N,r}|} \sum_{z\in I_{N,r}} w(z).$$

Proof. A period-R function has one independent value on each residue fiber, proving the first identity. Expanding

$$\sum_{I_{N,r}} |w(z)-\zeta|^2 = \sum_{I_{N,r}} |w(z)-\bar{w}_r|^2 + |I_{N,r}|\,|\zeta-\bar{w}_r|^2$$

proves the second. □

By Cauchy–Schwarz, the fiber-mean quadratic certificate gives the explicit upper bound

$$\sum_{I_N} |w-\rho| \le |I_N|^{1/2}\left(\sum_{r \bmod R}\sum_{I_{N,r}} |w-\bar{w}_r|^2\right)^{1/2}.$$

Thus the interface is finite and auditable whenever literal values of w are known.

4 Products, phases and precise limitations

Proposition 4.1 (Exact-period products)**.** *If $w=\prod_{j=1}^m w_j$ and every w_j is bounded periodic of period R_j, then w is periodic of period* $\operatorname{lcm}(R_1,\dots,R_m)$*. It may be inserted directly in theorem 1.1 when*

$$q\operatorname{lcm}(R_1,\dots,R_m) \le (\log X)^{\eta_0}.$$

Rational additive phases fall in this proposition. A generic additive phase need not be well approximated by a small-period function. TPC-158 tests that boundary exactly. Likewise, theorem 2.2 treats one interval $(N,2N]$; it does not convert exceptional-scale density into control of predetermined prefix endpoints, the obstruction isolated in TPC-150.

5 Audit and progress classification

The accompanying standard-library audit source-locks the TPC-149 theorem and certificate using `CANONICAL_UTF8_LF_V2`, checks exact residue-fiber quadratic minimization on rational data, and checks the triangle decomposition on a signed fixture. Hashes serve only as integrity checks.

Statement	Status
Periodic-to-literal interface on the actual Möbius core	PROVED$_{L1}$
Production physical multiplier and approximation rate	NOT-TESTABLE
Generic phase or all prefixes	not proved
Positive fixed-X-power saving	not proved

6 Conclusion

The periodic theorem now has an exact literal-weight consumer: arithmetic cancellation is lost only through a normalized approximation cost whose unpenalized error term is residue-fiber-separable. This closes a genuine actual-core interface while exposing the next factual question: whether the eventual occurrence registry produces small cost. It does not answer that question by schema, numerics or notation.

References

[1] Terence Tao and Joni Teräväinen. Quantitative Correlations and Some Problems on Prime Factors of Consecutive Integers, 2026.